Homework/Programming Assignment #4 **Homework/midterm Due: 04/30/2026- 3:30PM**

Name/EID:
Email:
Signature (*required*)
I/We have followed the rules in completing this Assignment.

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Question	Points	Total
HA 1	50	
PA	100 +20 bonus	
Presentation* (Bonus)	20	

Instruction:

1. Remember that this is a graded assignment. It is the equivalent of a **midterm take-home exam**.
2. **You CANNOT use the extra 48-hour time budget on this assignment.**
3. *** You can present the results of the PA in the class time on April 30th** and receive extra bonus depending on the quality of your presentation!
4. **Teams who are interested in presenting their assignment should let me know by April 29th.**
5. **For PA questions**, you need to write a report showing how you derived your equations, describes your approach, test functions, and discusses the results. You should show your test results for each function.

3. You are to work **alone** or **in teams of two** and are **not to discuss the problems with anyone** other than the TAs or the instructor.
4. It is open book, notes, and web. But you should cite any references you consult.
5. Unless I say otherwise in class, it is due before the start of class on the due date mentioned in the P/H Assignment.
6. **Sign and append** this score sheet as the first sheet of your assignment.
7. Remember to submit your assignment in Canvas.

➢ Homework Assignment (HA)

1) In **W14-L1**, we derived admittance-based VFs for Tubular and cone types VFs. Extend these algorithms for Impedance-based VFs. Write appropriate equations and describe how you implement them in real applications.

➢ Programming Assignment (PA)

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Figure shows the four robots that you have selected in the beginning of semester and developed its forward kinematics and numerical inverse Kinematics algorithms in THA2.

For this PA, as shown in Fig. 1, you are going to use your selected robot with an attached cylindrical tool with 100 mm length and 5 mm diameter attached to its tool frame to implement the following constrained optimization-based control algorithms.

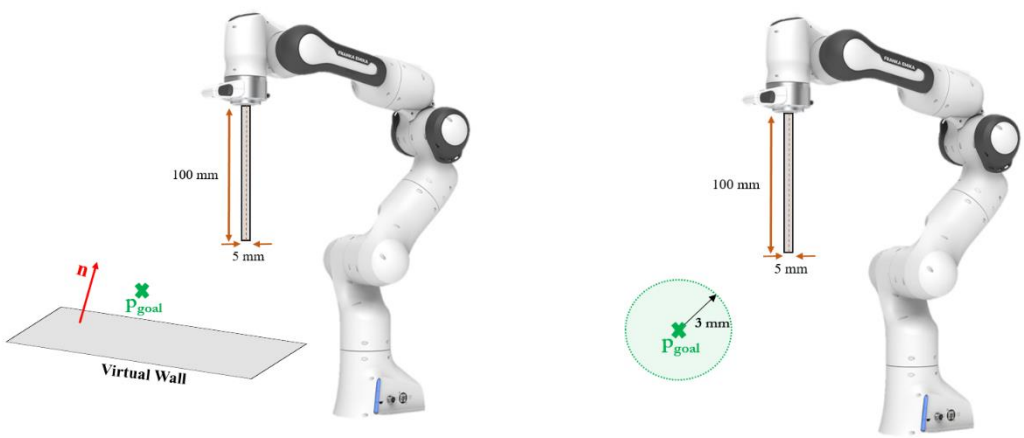

Fig. 1 Panda robot with an attached cylindrical tool to its tool frame

a) As shown in the right side of **Fig. 1** and similar to **Example 1 in W15-L1**, implement an algorithm that moves the tool tip toward an arbitrary goal point while

i) Keeps tool tip as close as possible to $\mathbf{p_{goal}}$ while never going beyond 3 mm from the goal
ii) and obeying joint limits.

b) As shown in the right side of **Fig. 1** and similar to **Example 1 in W15-L1**, implement an algorithm that moves the tool tip toward an arbitrary goal point while

i) Keeps tool tip as close as possible to $\mathbf{p_{goal}}$ while never going beyond 3 mm from the goal
ii) Obeying joint limits.
iii) minimizes the change in direction of the tool shaft during motion

c) **(+20 bonus)** As shown in the left side of **Fig. 1**, **repeat part (a) and (b) while having a virtual wall** defined close to the desired goal point. Note: make sure you define the virtual wall in a place that affects the trajectory of the robot obtained in part (a) and (b)

d) Compare performance of the robot motion in each case using appropriate plots, measures and for different configurations and locations for p_{goal} and virtual wall (if you are doing it).